

Process Characterization Plan

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCP-006

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Approvals	3
Abbreviations	3
1 Purpose and scope	5
2 Objectives	6
3 Related documents	7
4 Prior knowledge and quality risk basis	8
4.1 Unit-operation description and prior knowledge	8
4.2 Quality attributes in scope	9
4.3 Risk-based prioritization of parameters	10
5 Materials and methods	11
5.1 Scale-down model and its qualification	11
5.2 Operation	11
5.3 Analytical methods	12
5.4 Sampling plan	12
5.5 Statistical methods	13
6 Study design	14
6.1 Factors, ranges and study type	14
6.2 Screening design	15
6.3 Response-surface design	15
6.4 Univariate assessment	16
7 Acceptance and decision criteria	17
8 Proven acceptable ranges (planned analysis)	19
9 Data management and integrity	20
10 Roles and responsibilities	21
11 Deliverables and schedule	22
12 Risks and assumptions	23
13 Approvals	24
14 Appendices	25
14.1 Appendix A — Planned screening design matrix	25

14.2Appendix B — Planned response-surface design matrix	25
---	----

References 27

Field	Value
Document ID	PCP-006
Document class	Process Characterization Plan
Title	Process Characterization Plan: Low-pH Viral Inactivation (Step 6)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DoE, design

of experiments; GPP, general process parameter; HMW, high molecular weight; ICH, International Council for Harmonisation; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OLS, ordinary least squares; PAR, proven acceptable range; PRESS, predicted residual error sum of squares; RSM, response-surface methodology; SDM, scale-down model; SEC, size-exclusion chromatography; SOP, standard operating procedure; TCID₅₀, fifty per cent tissue culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

Low-pH viral inactivation is the first dedicated viral clearance step in the A-Mab purification train, and it is a hold and not a separation. The Protein A eluate is titrated to a low pH, held for a defined time at a controlled temperature, and then neutralised before cation exchange. The step carries the largest single contribution to the enveloped virus clearance claimed for the drug substance, and its operating pH is bounded on both sides, because inactivation becomes incomplete as the pH rises and the antibody aggregates as the pH falls.

This plan defines the process characterization study for the step, and it states what will be studied, at what scale, how the data will be analysed, and the criteria against which the executed study will be judged. It is written before any characterization data exist, so it records intentions and decision rules, and the results, the parameter classifications and the operating region will all be reported in PCR-006.

The study covers the parameters in Table 6.1 over the ranges given there, of which 3 (inactivation pH, hold time and temperature) will be studied in a multivariate design and 1 will be studied univariately (§6.4). The responses are the XMuLV log reduction factor, the aggregate level of the neutralised pool, and the acidic charge variant level of the same pool. Product quality runs and virus spiking runs will be executed on the same qualified scale-down model, within the framework of ICH Q5A(R2) (International Council for Harmonisation 2023a) and the lifecycle approach to process validation (U.S. Food and Drug Administration 2011).

Three limits on the scope are set here. Minute virus of mice is not inactivated at low pH, so this step is credited with enveloped virus clearance only, and the parvovirus claim rests on anion exchange and virus filtration (PCR-008 and PCR-009). Host cell protein precipitates during the hold in platform experience, but the extent varies between runs, so no host cell protein clearance will be claimed here and the Protein A pool level will be carried forward as the input to cation exchange (PCR-007). The step does not on its own satisfy the cumulative enveloped virus requirement, and the contribution required of it is derived in §7.

2 Objectives

The study has five objectives, and each is written as an intention, because no characterization data exist at the time of approval.

- (i) To quantify the effect of inactivation pH, hold time and temperature on XMuLV log reduction, on pool aggregate and on acidic charge variants, over the ranges in Table 6.1.
- (ii) To fit a response-surface model for each response and to define the multivariate operating region in which every acceptance criterion in Table 7.1 is met at the same time.
- (iii) To determine a proven acceptable range for each parameter against each attribute it governs, both at the set-point and with the other parameters varying within their normal operating ranges (§8).
- (iv) To generate the evidence needed to classify each parameter under SOP-4001, noting that the classifications themselves are assigned in PCR-006 and not in this plan.
- (v) To justify the Stage 2 operating ranges and in-process controls for the step, and to supply its clearance contribution to the cumulative viral safety claim consolidated in PCMR-001.

Objectives (i) to (iii) are discharged by the designs in §6 and the analyses in §7 and §8, whereas objectives (iv) and (v) depend on those outcomes and are discharged in the report.

3 Related documents

Table 3.1 lists the documents this plan depends on and the documents that depend on it: RA-001 sets the risk basis for the scope, PCMP-001 is the master plan over the campaign, and PCR-006 will report the executed study.

Table 3.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-006	Process Characterization Report (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2 lists the procedures and the validated analytical methods that govern execution and testing, and the reference numbers are controlled-document placeholders for this synthetic corpus.

Table 3.2: Governing procedures and validated analytical methods.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step is a hold, and its position in the train is fixed by the pH of the Protein A eluate. Table 4.1 gives the process train and the principal role of each step. The eluate leaves Protein A at a low pH already, so the titration to the inactivation set-point is small, and the pool is neutralised to the cation exchange load condition once the hold is complete. Mixing to pH homogeneity after acid addition and control of the hold temperature are the only physical demands the step makes on the vessel.

Table 4.1: The A-Mab drug-substance process train and the role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

Enveloped viruses are inactivated at low pH by irreversible disruption of the lipid envelope and of the envelope glycoproteins that mediate entry into the host cell. The mechanism acts on the virus and not on the antibody, and it does not apply to viruses that have no envelope, which is why the step is credited for XMuLV and makes no claim for MVM (§1).

Low-pH inactivation has been operated on the Novacyte humanised IgG1 platform for three earlier products (X-Mab, Y-Mab and Z-Mab), under conditions that have not changed materially between them. Prior knowledge from those products applies

directly to A-Mab for the inactivation kinetics, because the mechanism is a property of the virus and is independent of the product. It does not apply to the acid stability of the molecule, which is product specific, so the lower pH edge has to be established for A-Mab in this study.

Three mechanistic expectations follow from the platform data, and they frame the design. Inactivation should be faster at lower pH, at longer hold time and at higher temperature. Aggregate formation should rise with hold time and with temperature, and should rise steeply as the pH approaches the acid-unfolding transition of the antibody, while acidic charge variants should increase slightly across the hold without approaching the drug-substance limit. The study will confirm or refute each of these expectations.

4.2 Quality attributes in scope

The step sets one critical quality attribute and puts two others at risk, and Table 4.2 gives the attributes in scope with their acceptance criteria and their assigned criticality.

Table 4.2: Quality attributes in scope for the low-pH viral inactivation step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

Enveloped virus clearance is of very high criticality and is the attribute the step exists to deliver. Aggregate is of high criticality and is formed at this step and not cleared by it, so the hold has to be shown not to raise aggregate beyond the drug-substance limit. Acidic charge variants are of very low criticality and are monitored as a guard on acid exposure. Criticality follows the assessment recorded in RA-001, in which the Tool #1 score is the product of clinical impact and uncertainty, attributes scored high or very high are treated as critical, and criticality is treated as a continuum and not as a binary label (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009).

Enveloped virus clearance is cumulative across the process and is not delivered here alone. Anion exchange and small-virus retentive filtration each contribute an independent reduction by a different mechanism, and those contributions are characterized in PCR-008 and PCR-009, with the cumulative claim consolidated in PCMR-001. The contribution required of this step is the cumulative requirement less the clearance credited to the other two steps, and §7 gives the resulting criterion.

4.3 Risk-based prioritization of parameters

RA-001 ranked the parameters of this step on their potential effect on the attributes in Table 4.2 and on their potential to interact with each other, and the 3 highest ranked parameters were assigned to a multivariate study while A-Mab concentration was assigned to a univariate study. Table 6.1 carries that assignment.

Inactivation pH ranks highest on both scales. It drives the inactivation kinetics directly, it drives acid-induced aggregation directly, and the two effects run in opposite directions, so its acceptable range cannot be set from either response on its own. That two-sided dependence is why pH receives the most stringent treatment of any parameter in the step. Hold time and temperature act on the same kinetics as pH, which is where the interaction risk lies, since a short hold may be sufficient at the low end of the pH range and insufficient at the high end. Temperature acts on aggregation as well, so it appears in both mechanisms.

A-Mab concentration ranked low on both scales. Virus inactivation at fixed pH is a solution-phase reaction whose rate does not depend on the antibody concentration over the range in Table 6.1, and platform data show little or no effect on aggregate across the same range. A fourth factor would also expand a design that has to be executed under containment with spiked virus, at a cost the risk ranking does not justify. The parameter will still be studied across its full range (§6.4) and will still be classified in PCR-006, whatever the outcome.

5 Materials and methods

5.1 Scale-down model and its qualification

The scale-down model is a jacketed hold vessel with overhead mixing, titration by controlled acid addition, and continuous recording of pH and temperature. The step has no scale-dependent separation mechanism, so the model is matched to the commercial step on the variables that govern the outcome: the acid addition profile, the time to pH homogeneity after addition, the hold temperature and its control tolerance, and the neutralisation profile. Volume is not itself a scaling variable here, because inactivation is a solution-phase reaction in a homogeneous vessel.

The model will be qualified under SOP-1001 before any characterization run is executed. Qualification will compare the model against at-scale platform records for time to pH homogeneity, hold temperature profile, pool aggregate at the end of the hold, and the condition of the neutralised pool. The qualification record will be summarised in PCR-006 and filed against SOP-1001, and characterization runs will not begin until the model has been qualified, because every conclusion of this study is applied directly to the commercial-scale step.

Two configurations of the same model will be used. Unspiked runs generate the product-quality responses, and spiked runs generate the XMuLV log reduction factor and are executed in a containment laboratory, on material drawn from the same Protein A pool as the matching unspiked run. ICH Q5A(R2) requires that the spiked small-scale model represents the manufacturing step and that the spike does not itself alter it (International Council for Harmonisation 2023a). The volume of virus stock added will therefore be held to a small fraction of the pool volume, and the pH and conductivity of the pool will be recorded before and after spiking.

Centre-point runs are replicated in both designs (§6), and their variation is the pure-error term against which lack of fit will be tested, as well as a measure of the reproducibility of the whole procedure, from titration through to assay.

5.2 Operation

Each run will follow SOP-2009. The Protein A pool is adjusted to the target inactivation pH by addition of acid with mixing, the pH is confirmed on an independent probe, and the hold starts once the target is reached. The pool is held at the target temperature for the specified time, then neutralised to the cation exchange load pH with base, and filtered.

The input to the step is the Protein A pool characterized in PCR-005. Its aggregate level, protein concentration and pH are feed attributes of this step and not parameters of it. Buffer identity, acid and base molarity, filter type and vessel material are controlled to platform specification and are not characterized here. A-Mab concentration is the one feed attribute that will be varied deliberately (§6.4).

5.3 Analytical methods

Table 5.1 lists the validated methods that will generate the responses. XMuLV infectivity will be measured as a TCID50 titre under AMV-3017, and the log reduction factor is the difference between the titre of the spiked load and the titre of the held pool, each corrected for volume. Size variants will be measured by SEC under AMV-3011. Charge variants will be measured by imaged capillary isoelectric focusing under AMV-3013.

Table 5.1: Validated analytical methods for the study responses.

Reference	Title	Type
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

The precision of the infectivity assay bounds what the design can resolve. AMV-3017 reports an intermediate precision of 0.3 log₁₀ and a quantitation limit of 1.5 log₁₀ TCID50/mL, and the assay accounts for 28.0% of the observed variance in the response. Differences in log reduction smaller than the assay precision will not be treated as real, whatever their p value, and full method performance is held in the validation records listed in Table 3.2.

5.4 Sampling plan

Unspiked runs will be sampled at the end of the hold, after neutralisation and filtration, for aggregate and for charge variants, and a sample of the Protein A pool will be retained from each run, so that the change across the step can be reported and not only the outlet level.

Spiked runs will be sampled at the start and at the end of the hold for infectivity. Intermediate samples will be taken during the hold, so that the inactivation kinetics can be described and so that a run reaching the quantitation limit before the end of the hold can be identified as such. Cytotoxicity and interference controls will be run at the dilutions used for the samples, as ICH Q5A(R2) requires (International Council for Harmonisation 2023a).

Centre-point replication differs between the two designs, since the screening design carries 3 centre points and the response-surface design carries 4. Every centre point

is a complete run, executed from titration through to assay, so the replicates capture the variability of the procedure and not only the variability of the assay.

5.5 Statistical methods

Analysis will follow SOP-1002, and each factor is coded so that the edges of its characterization range are -1 and $+1$. The coded centre is the midpoint of that range, and it does not coincide with the set-point for any of the multivariate parameters, so predictions at the set-point will be made at the coded values of the set-points given in Table 6.1.

The screening design will be analysed by ordinary least squares with main effects and two-factor interactions. An effect is reported as twice the coded coefficient, so it is the change in the response across the whole studied range of that factor, and significance will be judged at $\alpha = 0.05$.

The screening analysis identifies which factors matter and how they interact, but it is not the predictive model. The response-surface model is the predictive model, and it is the basis of the operating region (§7) and of the proven acceptable ranges (§8). The response-surface design will be analysed as a full quadratic model in the same factors.

Model adequacy will be judged on R^2 , on adjusted R^2 , on predicted R^2 from the PRESS statistic, on the model F test, and on an analysis of variance that partitions the residual variation into lack of fit and pure error. §7 states what follows from each outcome.

Capability at commercial scale will be estimated by Monte-Carlo simulation of 2,000 batches, with each parameter drawn across its normal operating range and the fitted models used to predict the responses, and capability against a one-sided criterion will be reported as Cpk. A response with no active factor over the ranges studied will be reported as robust, and its factors will be retained in the knowledge space as evidence of that robustness.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters, their set-points, the ranges to be studied, the normal operating ranges and the study type, and Table 6.2 gives the coded letters used for the multivariate factors in the design matrices in Appendix A and Appendix B.

Table 6.1: Parameters, ranges and study type for the planned characterization. No classification is assigned in this plan.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Inactivation pH	pH	3.5	3.2-4	3.4-3.6	multivariate
Hold time	min	90	60-180	60-120	multivariate
Temperature	°C	21	15-25	19-23	multivariate
A-Mab concentration	g/L	20	5-35	5-35	univariate

Table 6.2: Coded factor legend for the planned design matrices.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

Every studied range is wider than the corresponding normal operating range, because the purpose of the study is to locate the edges and not to confirm the centre. The rationale for each range is given below.

The pH range, 3.2-4, spans both failure modes. Its lower edge sits just above the pH at which platform experience shows visible antibody precipitation, so the study can approach the aggregation edge and still produce material that can be assayed. Its upper edge sits above the pH at which enveloped virus inactivation is expected to become incomplete within the hold times studied. The normal operating range, 3.4-3.6, is narrow and lies well inside both edges. The set-point is pH 3.5.

The hold time range, 60-180 min, starts at the shortest hold that will be permitted at commercial scale and extends well beyond the set-point, so that extended holds caused by equipment or scheduling delays fall inside characterized territory. The upper edge

is the condition of greatest acid exposure, and so the worst case for aggregate and for charge variants.

The temperature range, 15–25 °C, spans the ambient conditions of the manufacturing suite and brackets the control tolerance of the hold vessel. The low edge is the worst case for inactivation and the high edge is the worst case for aggregate, so both edges of this range are informative.

The A-Mab concentration range, 5–35 g/L, covers the pool concentrations expected from Protein A across that step's own characterized load range (PCR-005). Its normal operating range and its studied range are the same, because the step imposes no control on the concentration it receives.

6.2 Screening design

The screening design is a two-level full factorial in the multivariate factors with 3 centre points, giving 11 runs in total, and the run order will be randomised. The design estimates all 8 factorial combinations, so every main effect and every two-factor interaction is estimated without aliasing, which is what the interaction risk identified in §4.3 requires. The planned coded matrix is given in Appendix A.

The screening design carries no axial points, so it cannot estimate curvature in any single factor. The centre points will detect curvature as a lack of fit, but they cannot attribute it to one factor, which is the reason for the response-surface design and the reason the screening effects will be reported as an identification of the active factors and not as a predictive model.

6.3 Response-surface design

The response-surface design is a face-centred central composite in the same factors, comprising 8 factorial points, 6 axial points and 4 centre replicates, giving 18 runs in total. The planned coded matrix is given in Appendix B.

The axial points are placed on the faces of the cube and not outside it, since a rotatable design would place them beyond the coded edges, which for pH would mean running below the precipitation threshold and above the pH at which inactivation is expected to fail. The face-centred form keeps every run inside the characterization ranges in Table 6.1, at the cost of a less precise estimate of the pure quadratic terms. That cost is accepted here, because the edges of the pH range are physically meaningful limits and not arbitrary ones.

6.4 Univariate assessment

A-Mab concentration will be studied one factor at a time, at the low edge, the set-point and the high edge of the range in Table 6.1, with pH, hold time and temperature held at their set-points. Both spiked and unspiked runs will be executed at each level.

The parameter is excluded from the multivariate design for the reasons given in §4.3, and the univariate result is informative either way. If no effect is found, the parameter will be classified accordingly in PCR-006 and its studied range will be retained in the knowledge space as evidence that the step is insensitive to it. If an effect is found, the multivariate design will be augmented before any operating region is declared.

7 Acceptance and decision criteria

Table 7.1 gives the acceptance criterion for each response together with the basis of that criterion. The aggregate and charge variant criteria are drug-substance specifications applied at the outlet of the step, and the viral criterion is derived, as set out below.

Table 7.1: Acceptance criteria for the study responses, with the basis of each.

Response	Acceptance criterion	Unit	Basis of the criterion
XMULV LRF (log ₁₀)	≥ 4.93	log ₁₀ (this step)	Step contribution back-calculated from the cumulative requirement
Aggregates (HMW, %)	≤ 5	% HMW (SEC)	Drug-substance specification, applied at the step outlet
Acidic variants	18–40	% (CEX/icIEF)	Drug-substance specification, applied at the step outlet

The enveloped virus criterion is a step contribution and not a cumulative figure. The drug-substance requirement is a cumulative XMULV reduction of at least 16.7 log₁₀ across the process, and anion exchange and small-virus retentive filtration are credited with 11.77 log₁₀ between them under the modular clearance approach, in which each step is spiked and evaluated independently of the others (International Council for Harmonisation 2023a; CMC Biotech Working Group 2009). This step must therefore achieve at least 4.93 log₁₀ of XMULV reduction across the operating region.

The clearance claim will be framed conservatively in three ways. The claim made for the step will be the reduction demonstrated at the worst case of the operating region, which is the combination of the highest pH, the shortest hold and the lowest temperature, and not the mean at the set-point. A run in which the held pool falls to the quantitation limit of AMV-3017 will be reported as a limit value, and the reduction will not be extrapolated beyond it. Inactivation will not be extrapolated to hold times shorter than the shortest studied, because the kinetics are steepest there and a linear extrapolation would overstate the clearance.

The product-quality criteria apply to the neutralised pool, where aggregate must be at or below 5 % HMW and acidic charge variants must lie between 18 % and 40 %. Both are drug-substance specifications. Applying them at the outlet of this step is deliberately conservative, because cation exchange reduces aggregate downstream of it (PCR-007). A run that exceeds either limit defines an edge of failure and will be reported as one.

Model adequacy decides what the operating region may be built on. A response whose response-surface model shows a significant lack of fit against pure error at $\alpha = 0.05$ will not be used to define the region, and the region will then be set from the responses whose models are adequate, together with the univariate data, with the shortfall stated plainly in PCR-006. A response with no significant factor effect will be reported as robust over the ranges studied, and its acceptance will be shown from the observed values and not from a model.

The operating region will be declared as the set of parameter combinations at which every criterion in Table 7.1 is met at the same time. It will be evaluated from the fitted response-surface models with the remaining parameters varying within their normal operating ranges, and not from the mean prediction alone. A model of mean response gives roughly even odds of meeting a limit at its own boundary, so a region drawn on the mean surface would not support the assurance that the viral criterion requires.

The pH criterion is two-sided, and it is the decisive one for this step, with its upper edge set by the log reduction criterion and its lower edge by the aggregate criterion. The study passes on pH only if those two edges leave an interval that contains the set-point and the whole normal operating range in Table 6.1, with margin on both sides. If they do not, no operating range for pH will be declared from this data set, the set-point and the normal operating range will be reset under change control, and the affected part of the study will be re-executed.

The study will be accepted as executed when four conditions hold. Every designed run in Appendix A and Appendix B has been executed, or its omission has been justified and assessed for impact. The centre-point variation is consistent with the precision of the methods in Table 5.1, and every spiked run carries passing cytotoxicity and interference controls. Any departure from this plan has been recorded, investigated, assessed for its effect on the conclusions and reported in PCR-006.

8 Proven acceptable ranges (planned analysis)

The operating region is multivariate, and it will be complemented by a proven acceptable range for each parameter against each attribute that parameter governs. The acceptance basis is the one already given in Table 7.1: aggregate and acidic charge variants are judged against their drug-substance specifications, and XMuLV reduction is judged against the step contribution derived in §7.

Two analyses will be run for each pairing of parameter and attribute. The first holds the other parameters at their set-points and scans the parameter of interest across its characterization range. The second varies the other parameters within their normal operating ranges, by Monte-Carlo sampling of the fitted response-surface model, and requires the criterion to be met across that variation, which makes it the reported default, because it is the condition under which the step will actually be operated.

The decision rule is the same for both analyses. The proven acceptable range is the contiguous interval of the parameter, inside its characterization range, that contains the set-point and over which the attribute stays within acceptance. Where that interval covers the whole characterization range, the parameter is proven acceptable across everything that was studied, and no claim will be made beyond it. Where the propagated analysis gives a narrower interval than the set-point analysis, the attribute that binds it will be named, together with the margin that remains to the set-point.

Each governed attribute will be reported with a table of the characterization range and both proven acceptable ranges, for every parameter, and with a two-panel figure for the parameter that governs it. The panels carry the set-point analysis and the propagated analysis, and each shows the parameter on the horizontal axis and the response on the vertical axis, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked.

A proven acceptable range and the operating region are not the same claim, and PCR-006 will keep them apart. A proven acceptable range is a statement about one parameter, and the operating region is a statement about the parameters together. Where every parameter turns out to be proven acceptable across its whole characterization range, the binding constraint on operation is the corner of the multivariate region and not any single range. Both are bounded by the ranges in Table 6.1 and by the fitted models, and neither will be claimed outside them.

9 Data management and integrity

Data will be recorded and retained in accordance with ALCOA+ principles. Run execution will be recorded in the electronic laboratory notebook against SOP-2009, with process data captured from the vessel control system, and analytical results will be recorded in the laboratory information management system under the method identifiers in Table 5.1.

Raw data, the fitted models and the analysis scripts will be retained together, so that every reported value can be traced back to the run that produced it. The statistical analysis will be reviewed independently by Statistics before the report is issued, and any change to this plan after approval requires a documented amendment approved by the functions in Table 1.

10 Roles and responsibilities

Table 10.1 assigns the functions involved in the study. MSAT owns this plan and the report, Statistics owns the design and the analysis and is accountable for the operating region and the proven acceptable ranges, and Quality Assurance approves the plan before execution begins.

Table 10.1: Roles and responsibilities for the planned study (synthetic).

Function	Responsibility
Manufacturing Science and Technology	Owns the plan and the report; qualifies the scale-down model
Process Development	Executes the unspiked characterization runs and the univariate assessment
Contract virology laboratory	Executes the spiked runs under Novacyte oversight
Analytical Development	Releases analytical data against the validated methods
Statistics	Owns the design, the analysis and the model adequacy assessment
Quality Assurance	Approves the plan, the deviations and the report

11 Deliverables and schedule

The deliverable of this plan is PCR-006, and Table 11.1 gives the sequence of work. No calendar dates are fixed here, and each phase begins once the preceding phase has been reviewed and released.

Table 11.1: Planned phase sequence for the study.

Phase	Content	Prerequisite
1	Scale-down model qualification against at-scale platform records	Approval of this plan
2	Screening design, unspiked and spiked	Qualified scale-down model
3	Response-surface design, unspiked and spiked	Screening result reviewed
4	Univariate assessment of A-Mab concentration	Qualified scale-down model
5	Analysis, operating region and proven acceptable ranges	Phases 2 to 4 complete
6	Issue of PCR-006	Analysis reviewed by Statistics

Scale-down model qualification precedes every characterization run. Screening precedes the response-surface study, because the response-surface factors are confirmed by the screening result. The univariate assessment may run in parallel with either. Within each design the spiked runs follow the matching unspiked runs, so that a run failing a product-quality criterion is known before virus is committed to it.

12 Risks and assumptions

Four risks to the study are recognised, and each carries a mitigation that is part of the plan.

The scale-down model may not represent the commercial hold. The step has no separation mechanism, which makes the model simpler to justify than a chromatography model, but titration and mixing are scale dependent, and the model will be qualified on those variables before execution (§5.1).

The spiked model may not represent the unspiked step. Virus stock carries cell culture matrix into the pool, and at high spike volumes it can shift the pH or the conductivity of the pool, so the spike volume will be held small and the pool condition will be recorded before and after spiking.

The infectivity assay may lose resolution to matrix cytotoxicity. A low-pH antibody pool is cytotoxic to the indicator cells at low dilution, and the higher dilution needed to avoid this costs resolution at the top of the reduction range. Cytotoxicity controls form part of every spiked run. A run that loses resolution will be reported as a limit value (§7).

The feed may not span the states the step will meet at scale. The input is the Protein A pool, whose own characterization is reported in PCR-005, and only its protein concentration is varied deliberately here, so feed states beyond that variation are covered by the input controls agreed with PCR-005 and not by this study.

Three assumptions are made explicit. The inactivation mechanism is a property of the virus envelope and transfers from the platform products to A-Mab, whereas the acid stability of the antibody does not transfer and is characterized here (§4.1). XMuLV is a conservative model virus for the enveloped viruses of concern under ICH Q5A(R2) (International Council for Harmonisation 2023a). The neutralised pool is the input to cation exchange, and no host cell protein clearance is credited to this step (§1).

13 Approvals

This plan takes effect once it has been approved by the functions listed in Table 1, and execution may not begin before that approval. Amendments after approval follow the change control route described in §9 and are approved by the same functions.

14 Appendices

The two planned design matrices follow, in coded form, and neither carries response columns, because no data exist at the time of approval.

14.1 Appendix A — Planned screening design matrix

Table 14.1: Planned screening design matrix, coded. Factor letters are given in Table 6.2.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	center	0	0	0
10	center	0	0	0
11	center	0	0	0

14.2 Appendix B — Planned response-surface design matrix

Table 14.2: Planned face-centred central composite design matrix, coded. Factor letters are given in Table 6.2.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1

Run	Type	A	B	C
8	factorial	1	1	1
9	axial	-1	0	0
10	axial	1	0	0
11	axial	0	-1	0
12	axial	0	1	0
13	axial	0	0	-1
14	axial	0	0	1
15	center	0	0	0
16	center	0	0	0
17	center	0	0	0
18	center	0	0	0

The same design in natural units is given below, for use as the run sheet.

Table 14.3: Planned response-surface design matrix in natural units.

Run	Type	Inactivation pH	Hold time	Temperature
1	factorial	3.2	60	15
2	factorial	3.2	60	25
3	factorial	3.2	180	15
4	factorial	3.2	180	25
5	factorial	4	60	15
6	factorial	4	60	25
7	factorial	4	180	15
8	factorial	4	180	25
9	axial	3.2	120	20
10	axial	4	120	20
11	axial	3.6	60	20
12	axial	3.6	180	20
13	axial	3.6	120	15
14	axial	3.6	120	25
15	center	3.6	120	20
16	center	3.6	120	20
17	center	3.6	120	20
18	center	3.6	120	20

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.